

```
1  # Name: Christopher A. Lee
2  # Date: 04/22/2026
3  # Prof: Dr. Callura
4  # Course: Bioinformatics – ENBC 455
5
6  # Libraries
7  from Bio import SeqIO
8  from collections import defaultdict
9
10 # Problem 1
11 with open("hw10_1.txt", "r") as f:
12     reads = [line.strip() for line in f.readlines() if line.strip()]
13     adj = {}
14     for read in reads:
15         prefix = read[:-1]
16         suffix = read[1:]
17         adj[prefix] = suffix
18
19     current_node = reads[0][:-1]
20     superstring = []
21
22     for i in range(len(reads)):
23         superstring.append(current_node[0])
24         current_node = adj[current_node]
25
26     print("".join(superstring))
```

TCCCTTCGGATTGCCTCTCTACTTACCTATGGTCCTTCAGTATACTTGATCCAGTCTGGCAAGTCTGCATTATCTCACCAGACTATGTCAAGCAGTTTGGCGACGTGGAATAACC
CTCCTCAAAGATGTAAGAAGCGAGATAAGTGGATTAGAGCGGCTGCATCGTACCTTTTGTAGCTTCCCGTCAATTCTGGTTCCAACCTACCCCAATTTGCAGACTTGTAAGGGT
CACCCAATAGGCAGAATGCGCATTCTTGGCTGTCTTACTTCAAAGACACACTCAGGGGGCGCCGCTGTATGGCGTTAGCATCGTCACCACAATCGGCCATTTAAGGGTGACGCTG
GGACATACGCAACAAAGATGCCGGGGCCGTCAGTTCCACGTAAGTTACTGCTAGAACTTTCTGTTGGACAAACCAATGAAATCCATGCGCTCTTGCTGGAATGTCATTACCCCC
GTTTAGTTGACGCTAACAACTTCTTGTAACGTTCTCGGAAGGGACTAGCGCTAAAGATGTTGCGTAGGACACCACCATTAGGTAACGGATCCTTCATTTACTCAGGGGCGTTGC
GCAACGAGGTCCCTATACTGTTCAAGGAAAGTATGTGCCAGCAGCCACATGCTGCCTGATAAGGATTCTCATACTGTGACGCGTGGCCTCGATGACCGTGAACAACCTAACCGTG
CTGAAATCGACACTTGCAATAGTTCGGACCGATGTAACATATTCCAACAGTGATATGTCATGTGGATGACTGCCAAGGAAAATAACAATATATCAAATCTCACTGCCATCCC
TGTACAATAGTACAGATCCTGGAAGTATCATATCACATCCCATAACCTCTATAAGTCTTATTTAATCTAAGTGGCCGATCGAAAGCTACCGCGAAAGGTTGACCTTCGCCTATC
AGGCATGTGTCAATAGGATACTGCACCACAGGCTCGGTAATAGTGCTCGTATCGCGTGCCCTTCCCAGATGAGGTAACACGCAGCGGAGTTACCCGGTTCGTCGCGTCGGGGCC
GAGATGACAAACCAAGCGCAGCGAGAGCACGCAGCACGTTTGCTACTAACACGTAAACGGTATCGGGACCAATTTTCGGAGTACCCTCGAAGCTTAGCTCGGCCTCCGCATGATC
ACTGATTCCACCGTCTATATTATCAGCGAAATTCAATCGCGCGCGGTTTCGATTATATCTGGACAATGAGCCGTTCTTTGCGATCTCACCTGCTCTAGAGTGCTCAACGGCCC
ACCATCCATGGTCGCTAAAGTAATTTCCCAAACCAATGTCCTAACTTAGGTTCAAGTGTCGAATTGTTGCCGTCCAGGGAGCGGTAGCTTGTGTGTCCTGCCATGCGGACTAGG
ATGACCTGTAAGGAGGAGCTTCGCCCCTAGCAATCCCCAACGGACGTAGCATCCGATTAGATTCATTCGCCGGACCTACGTGGGAGACCGACCATGTAGGGTAACCCCATGA
ACGGCTACGACTAAGATATCCTGACGCACGGAGCTTGAAGGAGTCGATCGTCGCCACCGCCTGGCAATTACGTCGTTTTCGAGGTAATGGCTGATGGAGATAGTCCACTCTCATG
TTCTGCTGAAGGATCCTGAAACGACTTTTAGGTTGATGCCAACACTACCCGCGCCAAAAATACCAATATTGGTGGTGATAGTAGGCCGCTAGGCGGCTTGAGACAAAAAGCACT
GACACAACCGGCGACAAACATTCTATCCTGTTTATCCCGAAGCCTAGAACCCTTTTGGCTGTGGGATGCTCGCGTCTCTTTGTAGCTTTTGTATTGGAGGGGGTGTACATTCTCC
AATCAACTCTGGCAAGGCTTGACCTAACAACCACTCTCGGGTCGATCTGTCACGACAGTCCTGCTGAATTCGTTGTTTTAGTCATACAGTAGGGCCTGGTAACGAAATTTTGCA
ACGGTGGCCAATATCGGTCTTTTCAGTTTGAAGGTGTGTTTAAGTAGGCGGCGCTAAGTACT

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Mothur

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VCF/BCF

BED

Built-in Converters

6: Clair3 on data 5 and data 3: merged output

898 lines 10 columns, 18 comments

format vcf_bgzip database ? size 14.4 KB

Preview Visualize Details Edit

Chrom	Pos	ID	Ref	Alt	Qual	Filter	Info	Format	data
##fileformat=VCFv4.2									
##source=Clair3									
##clair3_version=1.0.10									
##cmdline=/home/galaxyproject/galaxy/dependencies/_conda/envs/_clair3@1.0.10/bin/run_clair3.sh --bam_fn=input_reads.bam --ref_fn=reference.fasta --platform=ont --model_path=/home/galaxyproject/galaxy/dependencies/_conda/envs/_clair3@1.0.10/bin/models/r941_prom_sup_g5014 --output=. --threads=1 --include_all_ctgs --snp_min_af=0.08 --indel_min_af=chunk_size=5000000									
##reference=/home/galaxyproject/galaxy/database/jobs_directory/000/564/working/reference.fasta									
##FILTER=<ID=PASS,Description="All filters passed">									
##FILTER=<ID=LowQual,Description="Low quality variant">									
##FILTER=<ID=RefCall,Description="Reference call">									
##INFO=<ID=P,Number=0,Type=Flag,Description="Result from pileup calling">									
##INFO=<ID=F,Number=0,Type=Flag,Description="Result from full-alignment calling">									
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">									
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">									
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads 1. with MQ below 5 or an user-specified threshold, or 2. selected by 'samtools view -F 2316', are filtered)">									
##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allelic depths for the ref and alt alleles in the order listed">									
##FORMAT=<ID=PL,Number=G,Type=Integer,Description="Normalized, Phred-scaled likelihoods for genotypes as defined in the VCF specification">									
##FORMAT=<ID=AF,Number=A,Type=Float,Description="Observed allele frequency in reads, for each ALT allele, in the same order as listed, or the REF allele for a RefCall">									
##contig=<ID=chr20,length=64444167>									
#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	SAMPL
chr20	60361	.	C	T	9.70	PASS	F	GT:GQ:DP:AD:AF	0/1:9:8:
chr20	60403	.	A	C	10.25	PASS	F	GT:GQ:DP:AD:AF	0/1:10:8:
chr20	60466	.	C	T	5.47	PASS	F	GT:GQ:DP:AD:AF	0/1:5:8:
chr20	60546	.	C	T	9.86	PASS	F	GT:GQ:DP:AD:AF	0/1:9:9:
chr20	60555	.	T	A	10.19	PASS	F	GT:GQ:DP:AD:AF	0/1:10:9:
chr20	60580	.	G	A	6.23	PASS	F	GT:GQ:DP:AD:AF	0/1:6:9:
chr20	60651	.	G	C	10.26	PASS	F	GT:GQ:DP:AD:AF	0/1:10:9:
chr20	60721	.	T	C	7.39	PASS	F	GT:GQ:DP:AD:AF	0/1:7:9:
chr20	60796	.	G	T	9.79	PASS	F	GT:GQ:DP:AD:AF	0/1:9:9:
chr20	60806	.	T	C	6.72	PASS	F	GT:GQ:DP:AD:AF	0/1:6:10:
chr20	60864	.	A	C	6.56	PASS	F	GT:GQ:DP:AD:AF	0/1:6:11:
chr20	60886	.	C	T	7.21	PASS	F	GT:GQ:DP:AD:AF	0/1:7:11:
chr20	60901	.	C	G	7.54	PASS	F	GT:GQ:DP:AD:AF	0/1:7:11:
chr20	60923	.	T	C	7.67	PASS	F	GT:GQ:DP:AD:AF	0/1:7:11:
chr20	60988	.	A	C	10.10	PASS	F	GT:GQ:DP:AD:AF	0/1:10:1

History

search datasets

ENBC455-HW10-Problem2

93.1 MB

6: Clair3 on data 5 and data 3: merged output

5: GRCh38_no_alt_chr20.fa

4: GRCh38_no_alt_chr20.fa.fa.i

3: HG003_chr20.bam

2: HG003_chr20.bam.bai

1: Platform.txt

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- FASTQ Quality Control
- SAM/BAM
- BED
- VCF/BCF
- Nanopore
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- Lift-Over
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- Fetch Sequences/Alignments
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- Variant Calling
- ChIP-seq
- RNA-seq
- Multiple Alignments
- Phenotype Association
- Evolution
- Regional Variation
- Chromosome Conformation
- Transposon Insertion Sequencing
- Protein Modeling
- Genome Editing
- Biodiversity data exploration
- Sequence Contamination Filtering
- Genome Diversity

4: DeepVariant on dataset 2 and 3: VCF file ok									
25,284 lines 10 columns, 23 comments									
format vcf database ? size 2.4 MB									
Preview Visualize Details Edit									
Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10
##fileformat=VCFv4.2									
##FILTER=<ID=PASS,Description="All filters passed">									
##FILTER=<ID=RefCall,Description="Genotyping model thinks this site is reference.">									
##FILTER=<ID=LowQual,Description="Confidence in this variant being real is below calling threshold.">									
##FILTER=<ID=NoCall,Description="Site has depth=0 resulting in no call.">									
##INFO=<ID=END,Number=1,Type=Integer,Description="End position (for use with symbolic alleles)">									
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">									
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Conditional genotype quality">									
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read depth">									
##FORMAT=<ID=MIN_DP,Number=1,Type=Integer,Description="Minimum DP observed within the GVCF block.">									
##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Read depth for each allele">									
##FORMAT=<ID=VAF,Number=A,Type=Float,Description="Variant allele fractions.">									
##FORMAT=<ID=PL,Number=G,Type=Integer,Description="Phred-scaled genotype likelihoods rounded to the closest integer">									
##FORMAT=<ID=PS,Number=1,Type=Integer,Description="Phase set">									
##FORMAT=<ID=MF,Number=R,Type=Float,Description="Methylation fraction for each of the reference and alternate allele">									
##FORMAT=<ID=MD,Number=R,Type=Integer,Description="Methylation depth for each of the reference and alternate allele">									
##FORMAT=<ID=MT,Number=1,Type=String,Description="Methylation type: 0/0=Unmethylated, 0/1=Heterozygous, 1/1=Methylated">									
##FORMAT=<ID=MI,Number=1,Type=Float,Description="Allele-specific methylation score: p-value for Wilcoxon Rank-Sum test based on the observed difference in methylation between haplotypes.">									
##FORMAT=<ID=MED_DP,Number=1,Type=Integer,Description="Median DP observed within the GVCF block rounded to the nearest integer.">									
##FORMAT=<ID=MID,Number=1,Type=String,Description="Identifies which model called this variant.">									
##DeepVariant_version=1.10.0									
##contig=<ID=chr20,length=64444167>									
#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	default
chr20	60014	.	AG	A	0	RefCall	.	GT:GQ:DP:AD:VAF:MID:PL	0/0:29:4:2,2:0.5:small_model
chr20	60279	.	A	C	0	RefCall	.	GT:GQ:DP:AD:VAF:MID:PL	0/0:37:7:4,2:0.285714:small_r
chr20	60343	.	G	C	0.1	NoCall	.	GT:GQ:DP:AD:VAF:MID:PL	./.:19:7:5,2:0.285714:deepvaria
chr20	60347	.	TC	T	0.3	NoCall	.	GT:GQ:DP:AD:VAF:MID:PL	./.:12:8:6,2:0.25:deepvariant:0,
chr20	60355	.	AC	A	0.1	NoCall	.	GT:GQ:DP:AD:VAF:MID:PL	./.:18:4:2,2:0.5:deepvariant:0,2
chr20	60359	.	CA	C	0	RefCall	.	GT:GQ:DP:AD:VAF:MID:PL	0/0:34:6:4,2:0.333333:smallL
chr20	60361	.	C	T	0	RefCall	.	GT:GQ:DP:AD:VAF:MID:PL	0/0:21:5:3,2:0.4:small_model:t
chr20	60368	.	CCA	C	0.1	NoCall	.	GT:GQ:DP:AD:VAF:MID:PL	./.:17:8:5,2:0.25:deepvariant:0,
chr20	60403	.	A	C	0	RefCall	.</		

History

search datasets

ENBC455-HW10Problem3

Default Storage

95.6 MB

4

4: DeepVariant on dataset 2 and 3: VCF file

Add Tags

25,284 lines 10 columns, 23 comments
format **vcf**, database ?

***** Intermediate results will be written to

1. Chrom

```
##fileformat=VCFv4.2
##FILTER=<ID=PASS,Description="All filter:
##FILTER=<ID=RefCall,Description="Genotyp
##FILTER=<ID=LowQual,Description="Confide
##FILTER=<ID=NoCall,Description="Site has
```

3: GRCh38_no_alt_chr20.fa

2: HG003_chr20.bam

1: Platform.txt